

M 16020 – EXAM FORMULAS
THE SECOND DERIVATIVE TEST

Suppose f is a function of two variables x and y , and that all the second-order partial derivatives are continuous. Let

$$d = f_{xx}f_{yy} - (f_{xy})^2$$

and suppose (a, b) is a critical point of f .

1. If $d(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a relative minimum at (a, b) .
2. If $d(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a relative maximum at (a, b) .
3. If $d(a, b) < 0$, then f has a saddle point at (a, b) .
4. If $d(a, b) = 0$, the test is inconclusive.

LAGRANGE EQUATIONS

For the function $f(x, y)$ subject to the constraint $g(x, y) = c$, the Lagrange equations are

$$f_x = \lambda g_x \quad f_y = \lambda g_y \quad g(x, y) = c$$

GEOMETRIC SERIES

If $0 < |r| < 1$, then

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$$

TAYLOR SERIES

The Taylor series of $f(x)$ about $x = c$ is the power series

$$\sum_{n=0}^{\infty} a_n (x - c)^n \quad \text{where} \quad a_n = \frac{f^{(n)}(c)}{n!}$$

Examples:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \text{ for } -\infty < x < \infty; \quad \ln x = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n, \text{ for } 0 < x \leq 2$$

VOLUME & SURF AREA

Right Circular Cylinder

$$V = \pi r^2 h$$

$$S = \begin{cases} 2\pi r^2 + 2\pi rh \\ \pi r^2 + 2\pi rh \end{cases}$$

Sphere

$$V = \frac{4}{3}\pi r^3$$

$$S = 4\pi r^2$$

Right Circular Cone

$$V = \frac{1}{3}\pi r^2 h$$

$$S = \pi r \sqrt{r^2 + h^2} + \pi r^2$$

Practice Problems

1. Evaluate $\int \frac{1}{(3x-1)^4} dx$.

A. $\frac{12}{(3x-1)^5} + C$

B. $\frac{1}{9(3x-1)^3} + C$

C. $\frac{1}{(3x-1)^3} + C$

D. $\frac{1}{3(3x-1)^3} + C$

E. $\frac{4}{(3x-1)^5} + C$

2. Evaluate $\int e^{3-2x} dx$.

A. $2e^{3-2x} + C$

B. $\frac{1}{2}e^{3-2x} + C$

C. $\frac{e^{4-2x}}{4-2x} + C$

D. $\frac{1}{3}e^{3-2x} + C$

E. $\frac{e^{3-2x}}{3-2x} + C$

3. Find a function f whose tangent line has slope $x\sqrt{5-x^2}$ for each value of x and whose graph passes through the point $(2,10)$.

A. $f(x) = \frac{1}{3}(5-x^2)^{3/2}$

B. $f(x) = \frac{2}{3}(5-x^2)^{3/2} + \frac{28}{3}$

C. $f(x) = \frac{1}{3}(5-x^2)^{3/2} + \frac{29}{3}$

D. $f(x) = \frac{1}{3}(5-x^2)^{3/2} + \frac{31}{3}$

E. $f(x) = \frac{3}{2}(5-x^2)^{3/2} + \frac{17}{2}$

4. Evaluate $\int x \ln(x^2) dx$.

A. $\frac{1}{2}x^2 \ln x^2 - \frac{1}{2}x^2 + C$

B. $\frac{1}{2}x^2 \ln x^2 - \frac{1}{2}x + C$

C. $\frac{1}{2}x^2 \ln x^2 - \frac{1}{6}x^3 + C$

D. $x \ln x^2 + \frac{1}{x} + C$

E. $\frac{1}{2}x^2 \ln x - \frac{1}{2}x^2 + C$

5. The area of the region bounded by the curves $y = x^2 + 1$ and $y = 3x + 5$ is

- . $\frac{125}{6}$
- B. $\frac{56}{3}$
- C. $\frac{27}{2}$
- D. $\frac{25}{6}$
- E. $\frac{32}{3}$

6. If $f(x, y) = (xy + 1)^2 - \sqrt{y^2 - x^2}$, evaluate $f(-2, 1)$.

- . 1
- B. $1 - \sqrt{5}$
- C. Not defined
- D. $1 - \sqrt{5}$
- E. $1 - \sqrt{3}$

7. A paint store carries two brands of latex paint. Sales figures indicate that if the first brand is sold for x_1 dollars per gallon and the second for x_2 dollars per gallon, the demand for the first brand will be $D_1(x_1, x_2) = 100 + 5x_1 - 10x_2$ gallons per month and the demand for the second brand will be $D_2(x_1, x_2) = 200 - 10x_1 + 15x_2$ gallons per month. Express the paint store's total monthly revenue, R , as a function of x_1 and x_2 .

- . $R = x_1 D_1(x_1, x_2) + x_2 D_2(x_1, x_2)$
- B. $R = D_1(x_1, x_2) + D_2(x_1, x_2)$
- C. $R = D_1(x_1, x_2) D_2(x_1, x_2)$
- D. $R = x_2 D_1(x_1, x_2) + x_1 D_2(x_1, x_2)$
- E. $R = x_1 x_2 + D_1(x_1, x_2) D_2(x_1, x_2)$

8. Compute $\frac{\partial z}{\partial x}$, where $z = \ln(xy)$.

- . $\frac{1}{x}$
- B. $\frac{1}{y}$
- C. $\frac{1}{xy}$
- D. $\frac{1}{x} + \frac{1}{y}$
- E. $\frac{y}{x}$

9. Compute f_{uv} if $f = uv + e^{u+2v}$.

- . 0
- B. $u + 2e^{u+2v}$
- C. $v + 2e^{u+2v}$
- D. $1 + 2e^{u+2v}$
- E. $1 + e^{u+2v}$

10. Find and classify the critical points of $f(x, y) = (x - 2)^2 + 2y^3 - 6y^2 - 18y + 7$.
- A. (2,3) saddle point; (2, -1) relative minimum
 - B. (2,3) relative maximum; (2, -1) relative minimum
 - C. (2,3) relative minimum; (2, -1) relative maximum
 - D. (2,3) relative maximum; (2, -1) saddle point
 - E. (2,3) relative minimum; (2, -1) saddle point
11. A manufacturer sells two brands of foot powder, brand A and brand B. When the price of A is x cents per can and the price of B is y cents per can the manufacturer sells $40 - 8x + 5y$ thousand cans of A and $50 + 9x - 7y$ thousand cans of B. The cost to produce A is 10 cents per can and the cost to produce B is 20 cents per can. Determine the selling price of brand A which will maximize the profit.
- A. 40 cents
 - B. 45 cents
 - C. 15 cents
 - D. 50 cents
 - E. 35 cents
12. Use increments to estimate the change in z at (1,3) if $\frac{\partial z}{\partial x} = 2x - 4$, $\frac{\partial z}{\partial y} = 2y + 7$, the change in x is 0.3 and the change in y is 0.5.
- A. 7.1
 - B. 2.9
 - C. 4.9
 - D. 5.9
 - E. 6.3
13. Using x worker-hours of skilled labor and y worker-hours of unskilled labor, a manufacturer can produce $f(x, y) = x^2y$ units. Currently 16 worker-hours of skilled labor and 32 worker-hours of unskilled labor are used. If the manufacturer increases the unskilled labor by 10 worker-hours, use calculus to estimate the corresponding change that the manufacturer should make in the level of skilled labor so that the total output will remain the same.
- A. Reduce by 4 hours.
 - B. Reduce by 10 hours.
 - C. Reduce by $\frac{5}{4}$ hours.
 - D. Reduce by $\frac{5}{2}$ hours.
 - E. Reduce by 5 hours.

14. Find the maximum value of the function $f(x, y) = 20x^{3/2}y$ subject to the constraint $x + y = 60$. Round your answer to the nearest integer.

. 84,654
B. 188,334
C. 4,320
D. 259,200
E. 103,680

15. Evaluate $\int_1^2 \int_0^1 (2x + y) \, dy \, dx$.

. $\frac{9}{2}$
B. $\frac{5}{2}$
C. $\frac{3}{2}$
D. $\frac{7}{2}$
E. $\frac{1}{2}$

16. The general solution of the differential equation $\frac{dy}{dx} = 2y + 1$ is:

. $x = y^2 + y + C$
B. $2y + 1 = Ce^{2x}$
C. $y = 2xy + x + C$
D. $y = Ce^{2x} - 2y - 1$
E. $y = Ce^{2x}$

17. The value, V , of a certain \$1500 IR account grows at a rate equal to 13.5% of its value. Its value after t years is:

. $V = 1500e^{-0.135t}$
B. $V = 1500 + 0.135t$
C. $V = 1500e^{0.135t}$
D. $V = 1500(1 + 0.135t)$
E. $V = 1500 \ln(0.135t)$

18. It is estimated that t years from now the population of a certain town will be increasing at a rate of $5 + 3t^{2/3}$ hundred people per year. If the population is presently 100,000, by how many people will the population increase over the next 8 years?

. 100
B. 9,760
C. 6,260
D. 24,760
E. 17,260

19. Calculate the improper integral $\int_0^{\infty} x e^{-x} dx$.

- A. $\frac{1}{2}$
- B. 1
- C. $\frac{1}{2}$
- D. $\frac{5}{2}$
- E. The integral diverges.

20. An object moves so that its velocity after t minutes is given by the formula $v = 20e^{-0.01t}$. The distance it travels during the 10th minute is

- A. $\int_0^{10} 20e^{-0.01t} dt$
- B. $\int_9^{10} (20e^{-0.01t}) dt$
- C. $\int_0^{10} (20e^{-0.01t}) dt$
- D. $\int_9^{10} 20e^{-0.01t} dt$
- E. $\int_9^{10} (0.2e^{-0.01t}) dt$

21. Find the sum of the series $\sum_{n=1}^{\infty} \left(\frac{2}{3} \right)^n$.

- A. $\frac{2}{5}$
- B. $\frac{2}{5}$
- C. $\frac{3}{2}$
- D. $\frac{3}{2}$
- E. The series diverges.

22. Use a Taylor polynomial of degree 2 to approximate $\int_0^{0.1} \frac{100}{x^2 + 1} dx$. Round your answer to five decimal places.

- A. 9.96687
- B. 10.00000
- C. 9.96677
- D. 9.66667
- E. 9.96667

23. Find the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{n3}{5} \frac{x}{+1}$.

- A. $\frac{5}{3}$
- B. 1
- C. $\frac{3}{25}$
- D. $\frac{3}{5}$
- E. ∞

24. Find the Taylor series of $f(x) = \frac{x}{2+x^2}$ at $x = 0$.

- A. $\sum_{n=0}^{\infty} \frac{x^{n+1}}{2^{n+1}}$
- B. $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2^n}$
- C. $\sum_{n=0}^{\infty} (-1)^{2n+1} x^{2n+1}$
- D. $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{2^{n+1}}$
- E. $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+1}}{2^{n+1}}$

25. Write the following infinite series in summation notation.

$$5 - \frac{7}{8} + \frac{9}{27} - \frac{11}{64} + \dots$$

- A. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n+5}{n^3}$
- B. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2n+3}{n^3}$
- C. $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{3n+2}{n^3}$
- D. $\sum_{n=1}^{\infty} (-1)^n \frac{2n+5}{2}$
- E. $\sum_{n=1}^{\infty} (-1)^n \frac{2n+3}{2}$

26. Determine which of the following series converge.

I. $\sum_{k=2}^{\infty} \frac{k^2}{5^k}$

II. $\sum_{k=3}^{\infty} \frac{(3k+1)\pi^{2k}}{10^{k+1}}$

III. $\sum_{k=1}^{\infty} \frac{k!}{\binom{k}{2}^k}$

- . III
 B. I & II
 C. I & III
 D. II & III
 E. II

27. Find the Taylor series about $x = 0$ for the indefinite integral

$$\int x e^{-x^3} dx.$$

- . $\sum_{n=0}^{\infty} -\frac{1}{(3^n+1)} x^{3^n+2} + C$
 B. $\sum_{n=0}^{\infty} -\frac{\binom{1}{n}}{(3^n+2)} x^{3^n+2} + C$
 C. $\sum_{n=0}^{\infty} -\frac{\binom{1}{n}}{(3^n+1)} x^{3^n+2} + C$
 D. $\sum_{n=0}^{\infty} -\frac{1}{(3^n+2)} x^{3^n+2} + C$
 E. $\sum_{n=0}^{\infty} -\frac{\binom{1}{n}}{(3^n+1)} x^{3^n+1} + C$

28. A patient is given an injection of 50 milligrams of a drug every 24 hours. After t days, the fraction of the drug remaining in the patient's body is

$$f(t) = 2^{-t/3}.$$

If the treatment is continued indefinitely, approximately how many milligrams of the drug will eventually be in the patient's body just prior to an injection?

- . 202.7
 B. 152.7
 C. 305.4
 D. 242.4
 E. 192.4

29. Compute $\int (\sin x - \cos x)(\sin x + \cos x)^5 dx$.

- A. $\frac{1}{6}(\cos x + \sin x)^6 + C$
- B. $6(\cos x + \sin x)^6 + C$
- C. $\frac{1}{6}(\sin x + \cos x)^6 + C$
- D. $6(\sin x + \cos x)^6 + C$
- E. $\frac{1}{6}(\sin x + \cos x)^6 + C$

30. Evaluate $\int x^2 \cos(5x) dx$.

- A. $\frac{1}{5}x^2 \sin(5x) + \frac{2}{25}x \cos(5x) + \frac{2}{125} \sin(5x) + C$
- B. $\frac{1}{5}x^2 \sin(5x) - \frac{2}{25}x \cos(5x) - \frac{2}{125} \sin(5x) + C$
- C. $5x^2 \sin(5x) + 50x \cos(5x) + 250 \sin(5x) + C$
- D. $5x^2 \cos(5x) - 50x \sin(5x) - 250 \cos(5x) + C$
- E. $5x^2 \sin(5x) - 50x \cos(5x) - 250 \sin(5x) + C$

31. Evaluate $\int_e^5 \frac{\ln(x^4)}{x} dx$.

- A. $\frac{1}{8}(25 - e^2)$
- B. $2(25 - e^2)$
- C. $2(\ln 5)^2 - 2$
- D. $\frac{1}{8}(\ln 5)^2 - \frac{1}{8}$
- E. $\ln(25) - 2$

32. Find the volume of the solid generated by revolving the region bounded by:

$$y = 3e^{2x}, y = 0, x = 1, \text{ and } x = 3$$

about the x-axis.

- A. $\frac{3\pi}{4}(e^8 - 1)e^4$
- B. $\frac{3\pi}{4}(e^8 - 1)e^2$
- C. $\frac{9\pi}{2}(e^4 - 1)e^2$
- D. $\frac{9\pi}{4}(e^8 - 1)e^4$
- E. $\frac{3\pi}{2}(e^4 - 1)e^2$

33. Find the volume of the solid which has square cross-sections with side length $5x^2$ at each point $2 \leq x \leq 4$.

- . $\int_2^4 5\pi x^2 dx$
- B. $\int_2^4 5x^2 dx$
- C. $\int_2^4 5x^4 dx$
- D. $\int_2^4 25\pi x^4 dx$
- E. $\int_2^4 25x^4 dx$

34. The velocity of a car over the time period $0 \leq t \leq 3$ is given by the function

$$v(t) = 60te^{\frac{t}{4}}$$

miles per hour, where t is time in **hours**. What was the distance the car traveled in the first 90 **minutes**? Round your answer to two decimal places.

- . 166.42 miles
- B. 156.19 miles
- C. 126.63 miles
- D. 75.85 miles
- E. 52.78 miles

35. Given that $f(x, y) = \tan(xy^3)$, compute $f_x(2\pi, \frac{1}{2})$.

- . $\frac{3}{2}$
- B. $\frac{\pi}{2}$
- C. 1
- D. 6π
- E. $\frac{1}{4}$

36. Let $h(x, y) = y \sin(xy)$. Find $\frac{\partial^2 h}{\partial y \partial x}$.

- . $2xy \sin(xy)$
- B. $2y \cos(xy) - xy^2 \sin(xy)$
- C. $y^3 \sin(xy)$
- D. $\cos(xy) + y^2 \sin(xy)$
- E. $(x + 1) \cos(xy) - x^2 y \sin(xy)$

37. A nature preserve wishes to construct a large compound which will hold both lions and gazelles. They currently have 6 gazelles. They estimate that if they use an area of x square miles and introduce L lions, then they will be able to support a population of G gazelles, given by the function

$$G(x, L) = 6 + 40x - 2x^2 - 18L^2 + 176L - 8xL$$

What conditions will lead to the largest number of gazelles?

- A. $L = 3, x = 5$
 - B. $L = 4, x = 4$
 - C. $L = 5, x = 4$
 - D. $L = 5, x = 3$
 - E. There are no such conditions because the function does not have a maximum.
38. Evaluate $\iint_R (e^x + 1) \, dA$, where R is the region indicated by the boundaries below:

$$0 \leq x \leq 1; \quad 0 \leq y \leq x$$

- A. 0
 - B. $\frac{1}{2}e$
 - C. $\frac{1}{2}e^2$
 - D. $\frac{1}{2}(e^2 - e)$
 - E. $e^2 - e$
39. Compute AB and BA , if possible, for the matrices:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 1 \\ 5 & 1 \\ 2 & 0 \end{bmatrix}$$

A. AB is not possible, and $BA = \begin{bmatrix} 1 & 3 \\ 11 & 3 \\ 4 & 0 \end{bmatrix}$

B. BA is not possible, and $B = \begin{bmatrix} 1 & 11 & 4 \\ 3 & 3 & 0 \end{bmatrix}$

C. BA is not possible, and $B = \begin{bmatrix} 0 & 3 \\ 10 & 2 \\ 4 & 2 \end{bmatrix}$

D. BA is not possible, and $B = \begin{bmatrix} 0 & 10 & 4 \\ 3 & 2 & 2 \end{bmatrix}$

E. Both AB and BA are not possible.

40. Find the general solution to the differential equation

$$x^5 \sin x + xy' = 3y, \quad x > 0$$

- A. $y = x \cos x - \sin x + C$
- B. $y = x \cos x + \sin x + C$
- C. $y = x \cos x + \sin x + C$
- D. $y = x^4 \cos x + x^3 \sin x + Cx^3$
- E. $y = x^4 \cos x + x^3 \sin x + Cx^3$

41. The amount of carbon, in grams, in a sample of soil is given by a function, $F(t)$, satisfying the differential equation:

$$F' + aF - b = 0$$

where a and b are constants, and time, t , is measured in years. If the sample originally contains 10 grams of carbon, which expression represents the amount of carbon present after 5 years?

- A. $\frac{b}{a} + (10 - \frac{b}{a})e^{5a}$
- B. $\frac{b}{a} + (10 - \frac{b}{a})e^{-5a}$
- C. $ab + (10 - ab)e^{-5a}$
- D. $ab + (10 - ab)e^{5a}$
- E. $\frac{b}{a} + 10e^{5a}$

42. Let $M = \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$. Compute $3M - M^2$.

- A. $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$
- B. $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$
- C. $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$
- D. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$
- E. $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$

Answers to Practice Problems

1. B
5.
9. D
13. D
17. C
21. B
25. B
29. C
33. E
37. B
41. B
~~45. B~~

2. B
6. C
10. E
14. E
18. B
22. E
26. B
30.
34. E
38. D
42. 
~~46. _____~~

3. D
7.
11.
15. D
19. C
23.
27. B
31. C
35. E
39. ~~C~~
~~43.~~
~~47. D _____~~

4.
8.
12. D
16. B
20. D
24. E
28. E
32. D
36. B
40. D
~~44. B~~
~~48. D _____~~